

Multiblock CAVI Contracts by a Gauss–Seidel Interaction Matrix

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Status and source question

This is a `candidate_full_likely_valid` answer to the multiblock extension left for future work by Rocco Caprio, Adrien Corenflos, and Sam Power, *Wasserstein Contraction of Coordinate Ascent Variational Inference*, arXiv:2605.30253 [1]. After their two-block contraction theorem, the authors ask for the dynamics when the number of blocks is $m > 2$ (source PDF page 5).

and Léonard, 2010). Moreover, it should be possible to consider multi-block algorithms, i.e. the setting where $m > 2$. We leave each of these extensions for future work. Our argument for coordinate ascent variational inference also suggests a novel convergence analysis for the standard coordinate descent algorithm under cross-smoothness and gradient quadratic growth conditions, which might be of independent interest; we give a more detailed discussion of this

The theorem below gives the systematic-scan multiblock extension under the natural matrix-valued version of the source’s Fisher-smoothness hypothesis. It also identifies the convergence rate with the spectral radius of a Gauss–Seidel comparison matrix.

Setting and proof intuition

Let $m \geq 2$, let $\mathcal{X}_i$ be smooth complete Riemannian manifolds, and let T_i be the i th CAVI optimality map: $T_i[\mu_{-i}]$ is the minimizing i th marginal when all other marginals are fixed. Fix a product fixed point $\mu^* = (\mu_1^*, \dots, \mu_m^*)$, so $\mu_i^* = T_i[\mu_{-i}^*]$.

Write W_i for the Wasserstein- p distance on $\mathcal{P}_p(\mathcal{X}_i)$ and $\mathcal{I}_i$ for relative Fisher information. Suppose μ_i^* satisfies

$$\lambda_i W_i(\eta_i, \mu_i^*) \leq \mathcal{I}_i(\eta_i \| \mu_i^*)^{1/2}. \quad (1)$$

The multiblock Fisher-smoothness matrix (L_{ij}) , with $L_{ii} = 0$, means that throughout a product neighborhood $\mathcal{N}$ of μ^* ,

$$\mathcal{I}_i(T_i[\eta_{-i}] \| T_i[\mu_{-i}^*])^{1/2} \leq \sum_{j \neq i} L_{ij} W_j(\eta_j, \mu_j^*). \quad (2)$$

For $m = 2$, this is exactly the source’s Fisher-smoothness condition. A stronger all-pairs version of (2) may of course be used.

The intuition is finite-dimensional. Combining (1) and (2) makes each new Wasserstein error at most a weighted sum of the other errors. In a sequential sweep, already-updated blocks occur on the left and not-yet-updated blocks on the right. Solving this triangular inequality produces the usual Gauss–Seidel matrix. Perron–Frobenius weighting then converts spectral radius below one into a contracting norm.

Main result

Put $a_{ij} = L_{ij}/\lambda_i$ for $i \neq j$ and $a_{ii} = 0$, and let $A = (a_{ij})$. Write A_- and A_+ for the strict lower and strict upper triangular parts of A , and define

$$G = (I - A_-)^{-1}A_+. \quad (3)$$

The inverse exists because A_- is nilpotent, and it is entrywise nonnegative.

Theorem 1 (systematic-scan multiblock contraction). *Assume (1) and (2). If*

$$\rho(G) < 1, \quad (4)$$

then for every $q \in (\rho(G), 1)$ there is a vector $w \in (0, \infty)^m$ such that one full systematic CAVI sweep satisfies

$$\max_i \frac{W_i(\mu_i^{k+1}, \mu_i^*)}{w_i} \leq q \max_i \frac{W_i(\mu_i^k, \mu_i^*)}{w_i}. \quad (5)$$

For a local statement, suppose $\mathcal{N} = \prod_i \{\eta_i : W_i(\eta_i, \mu_i^) < \varepsilon_i\}$. Then (5) and invariance of $\mathcal{N}$ hold whenever*

$$\max_i \frac{W_i(\mu_i^0, \mu_i^*)}{w_i} < R, \quad R < \min_i \frac{\varepsilon_i}{w_i}.$$

If the assumptions are global, the conclusion holds for every initialization.

Proof. One systematic sweep updates blocks in the order $1, \dots, m$:

$$\mu_i^{k+1} = T_i[\mu_1^{k+1}, \dots, \mu_{i-1}^{k+1}, \mu_{i+1}^k, \dots, \mu_m^k].$$

Set $e_i^k = W_i(\mu_i^k, \mu_i^*)$. Since $T_i[\mu_{-i}^*] = \mu_i^*$, first (1) and then (2) give

$$e_i^{k+1} \leq \sum_{j < i} a_{ij} e_j^{k+1} + \sum_{j > i} a_{ij} e_j^k. \quad (6)$$

In vector form,

$$e^{k+1} \leq A_- e^{k+1} + A_+ e^k.$$

Because $(I - A_-)^{-1} = I + A_- + \dots + A_-^{m-1}$ is nonnegative,

$$e^{k+1} \leq G e^k \quad (7)$$

componentwise.

Fix $q \in (\rho(G), 1)$ and let $\mathbf{1} = (1, \dots, 1)^\top$. The Neumann series

$$w = (I - G/q)^{-1} \mathbf{1} = \sum_{n=0}^{\infty} (G/q)^n \mathbf{1}$$

is finite and strictly positive. It satisfies $Gw = q(w - \mathbf{1}) \leq qw$. If $r = \max_i e_i^k/w_i$, then $e^k \leq rw$, and (7) yields $e^{k+1} \leq rGw \leq qrw$. This is (5).

For the local assertion, the assumed initial bound places every old block in $\mathcal{N}$. During the first sweep, apply (6) successively. The already-updated errors are bounded by the corresponding components of $Ge^0 \leq qRw$, while the remaining old errors are bounded by Rw ; hence all hybrid inputs stay in $\mathcal{N}$. The same argument closes the induction over sweeps and proves local invariance. $\square$

Corollary 2 (parallel/Jacobi update). *If all blocks are updated from the previous iterate in parallel, then $e^{k+1} \leq Ae^k$. Consequently $\rho(A) < 1$ gives the analogue of (5), with any $q \in (\rho(A), 1)$.*

Proof. There are no already-updated errors in (6), so the comparison matrix is A . Repeat the Neumann-series weighting argument. $\square$

Corollary 3 (recovery of the source theorem). *For $m = 2$,*

$$G = \begin{pmatrix} 0 & a_{12} \\ 0 & a_{21}a_{12} \end{pmatrix}, \quad \rho(G) = a_{12}a_{21} = \frac{L_{12}L_{21}}{\lambda_1\lambda_2}.$$

Thus (4) is precisely the source’s small-interaction condition and its one-sweep rate.

Verification, scope, and novelty status

The included script tests 25,000 random nonnegative interaction matrices, 100,000 error vectors, and 20,000 two-block reductions. It verifies the triangular recursion, the Neumann weight inequality, weighted contraction, and the exact two-block product formula. This is a regression check, not a substitute for the proof.

Eight focused proof passes are recorded in the accompanying attempt note. They replace an insufficient cycle-product guess by the full comparison matrix, distinguish systematic and parallel scans, establish local invariance, recover the two-block theorem exactly, and audit the spectral criterion against linear comparison dynamics.

The theorem assumes smooth blocks and a combined multiblock Fisher-smoothness bound of the form (2). It does not prove that arbitrary pairwise Fisher-information estimates tensorize into that bound, nor does it cover random-scan CAVI or the source’s one-nonsmooth-block data-augmentation refinement. These are separate extensions.

Cheap run indexes and bounded web/arXiv searches on 17 August 2026 used arXiv:2605.30253 and the phrases *multiblock Fisher-smoothness CAVI*, *transport-information multi-block variational inference*, and *Wasserstein contraction Gauss–Seidel CAVI*. They found the source and known multiblock results under strong log-concavity and block smoothness, but no extension based on a matrix of Fisher-smoothness constants and transport-information inequalities that gives direct Wasserstein contraction. Novelty confidence is moderate-high pending expert review, especially because the source is recent.

Human-review recommendation. Check that the intended application supplies the combined bound (2); conditional on it, the transport-information and matrix argument is self-contained. Also verify the desired scan convention before using G rather than A .

References

- [1] R. Caprio, A. Corenflos, and S. Power, *Wasserstein Contraction of Coordinate Ascent Variational Inference*, arXiv:2605.30253 (2026).